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Integrating fuzzy logic and neural networks with multi-criteria decision-making for intelligent evaluation of drug compound design attributes

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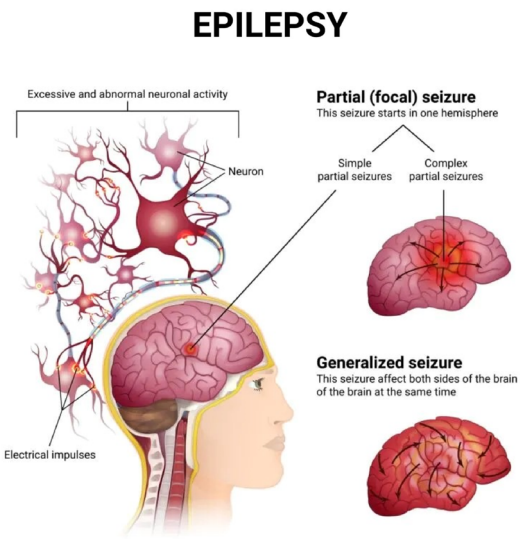
This study proposes a fuzzy machine learning framework for optimizing antiepileptic drug selection using Quantitative Structure-Property Relationship (QSPR) modeling under pharmacological uncertainty. Feature relevance was assessed using Random Forest-based importance and SelectKBest with mutual information, and a feedforward neural network was trained with 5-fold cross-validation. Fuzzy membership functions were incorporated to model variability in clinical and experimental data. Compared with Multiple Linear Regression and conventional QSPR models, the proposed approach achieved a 24 percent reduction in RMSE. Predicted pharmacological attributes were further integrated into a Multi-Criteria Decision-Making framework using TOPSIS to rank drug candidates based on efficacy, safety, and cost. The resulting rankings showed 95 percent Spearman correlation with clinician evaluations, demonstrating the framework reliability for uncertainty-aware antiepileptic drug prioritization and custom MCDM libraries for TOPSIS based prioritization.

Keywords Machine learning, Artificial neural networks, MCDM analysis, TOPSIS, Anti-epileptic drugs

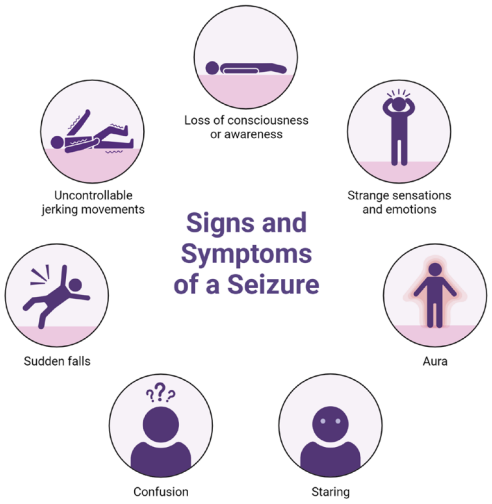
Epilepsy consists of a wide range of seizures due to chaotic electrical signals in the brain thereby interfering with sensations, behavior, muscle function and awareness[1]. An approximate amount of 50 million people in the world have epilepsy, of which 5 million new cases are reported annually. In the United States country wide, the number of adult patients suffering from epilepsy is approximately 3 million, and the number of children living with the condition is nearly 470,000[2]. Epileptic seizures, characterized by abnormal neuronal hyperactivity, are classified by the International League Against Epilepsy (ILAE) into focal, generalized and unknown onset types based on origin and awareness. A person with seizures experiences abnormal electrical brain activity and may have symptoms that include focal twitches, strange sensations or body rigidness and unconsciousness[3]. As well as shock and head movement, patients sometimes have absence seizures and confusion after seizures and these can be made worse by anxiety and memory issues as seen in Fig. 1. Seizure types are determined by clinical history, EEG recording and brain imaging prior to suggesting specific antiseizure treatments. AEDs, known as anticonvulsants, are medicines used to treat and control the frequency and strength of seizures in epilepsy[4]. Such medications work by altering neurotransmitter levels, adjusting the electrical properties of brain cells or controlling the activity of ion channels so that unusual electrical charges in the brain are prevented.

For Over time, both conventional and newer-generation AEDs have been developed, tailored to address different epilepsy syndromes with improved efficacy and reduced side effects. Some common AEDs are carbamazepine, valproic acid, lamotrigine, levetiracetam and topiramate, each with its own special effects and ways the body processes it[5]. Even though these drugs improve outcomes, it is essential to carefully consider which AED is best for a person and which is safest. Because decisions are often complex, it is helpful to rely on multi-criteria approaches when choosing treatment. Current research efforts are still concentrating on finding and perfecting new AEDs that can effectively manage seizures and cause the least harm to the mind or body[6].

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(a) Epilepsy Disease



(b) Sign and Symptoms of Epilepsy Disease

Fig. 1. Epilepsy and their Symptoms.

Machine Learning[7,8] is a branch of Artificial Intelligence (AI) that allows systems to find patterns in data, improve their performance and make predictions without being programmed. Machine Learning algorithms such as Artificial Neural Networks (ANNs) need precise, organized data to work on problems such as classification, regression and clustering. On the other hand, Fuzzy Machine Learning uses fuzzy logic concepts to address real world ambiguity and uncertainty[10]. With fuzzy set theory, Fuzzy Machine Learning makes it possible for a variable to be a member of many classes by a certain degree, instead of just one or the other[9]. This approach is particularly effective for handling noisy data, linguistic variables and scenarios with incomplete information[11]. Some applications include medical health settings depending on humans to assess symptoms and others require artificial intelligence to control systems like humans do. Fuzzy Artificial Neural Networks(FANNs) are hybrid systems that allow decisions to be taken with more precision, as the system balances probabilistic learning with interpretable fuzzy rules[12]. While machine learning works great for predictable tasks, fuzzy machine learning becomes important in difficult problem scenarios where some things are clear yet others are uncertain[13].

In discrete mathematics, graph theory is the core subject dedicated to graphs which represent connections between pairs of things[14]. A graph consists of vertices (nodes) connected by edges (links) and it is widely used for computer science, biology, chemistry, social networks and problems needing optimization. In traditional graph theory, vertices may be connected by an edge or not[15,16]. However, many real-world problems present unclear or uncertain information that cannot be well represented with crisp graphs. In order to overcome this difficulty, fuzzy graph theory was proposed as a specialized branch of classical graph theory. To show how vertices are

connected in a fuzzy graph, their links are given degrees of membership that lie between 0 and 1[17]. With this framework, complex systems can be represented more precisely which is useful in areas where having partial knowledge is common such as decision making, pattern recognition and data mining[18,19].

Molecular structures are described by topological indices, extracted from graph theory and employed frequently in mathematical chemistry and cheminformatics[20]. With these indices, we can measure the size, shape, branching and connection patterns of molecules without needing actual geometric or experimental data. A molecule is shown as a graph with each atom at a vertex and every bond between atoms presented by edges[21]. Well-known topological indices are Wiener index, Zagreb indices, Randic index and atom-bond connectivity index which identify different structural features. They play an important role in estimating the properties, activity and drug potential of compounds[22]. Topological indices are applied for fast virtual screening and structure activity relationship (SAR) investigations in drug discovery and the study of materials. By using machine learning, topological indices become important features in the construction of models for predictive computational chemistry[23].

MCDM relates to operations research where multiple possible options are assessed and measured when criteria are conflicting[24]. With MCDM, TOPSIS is one of the most preferred techniques due to its clear and logical way of working. In TOPSIS, the best choice should be the one with the shortest Euclidean distance to the ideal solution and the farthest one from the negative-ideal position. You normalize the matrix, calculate the weights for each factor and determine the distances from both the ideal and negative-ideal solutions[25]. It is particularly valuable when making important choices in medical diagnosis, supply management or when evaluating drugs because you have to weigh different factors at the same time[26,27]. The simplicity and efficiency of TOPSIS make it a valuable tool in supporting rational and transparent decision making[28].

Through QSPR analysis, numerical techniques are used to project the physicochemical and biological features of chemicals, including density (D), index of refractivity (IR), molar refractivity (MR), hbond acceptor (HBA), hbond donor (HBD), acd/logd (logD), acd/bcf (BCF), polarizability (POL), surface tension (ST) and molar volume (MV), based on their molecular structure[29]. Research analyzes eccentricity based topological indices of twenty epilepsy drugs namely Carbamazepine, Levetiracetam, Phenobarbital, Valproic acid, Ethosuximide, Zonisamide, Gabapentin, Pregabalin, Felbamate, Lacosamide, Tiagabine, Oxcarbazepine, Clonazepam, Eslicarbazepine, phenytoin, Fosphenytoin, Lamotrigine, Lacosamide, Lorazepam and Diazepam denoted as A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9, A_{10}, A_{11}, A_{12}, A_{13}, A_{14}, A_{15}, A_{16}, A_{17}, A_{18}, A_{19}, A_{20} through the use of QSPR analysis presented in Fig. 3. By finding a link between the model properties topological indices, geometrical factors, electronic characteristics and hydrophobicity with actual data on boiling point, solubility, toxicity or biological effects in experiments. QSPR plays an important role in drug discovery, material science and environmental chemistry[30], as illustrated in the Workflow in Fig. 2.

In the process, you identify appropriate descriptors, apply normalization or scaling on the data, choose machine learning algorithms and check the model's results with R^2, RMSE and cross-validation activities[31]. By using QSPR models, it is possible to pick out the most favorable substances from large groups of compounds without requiring many experiments in a laboratory. Model accuracy and reliability can only be achieved in cheminformatics and similar areas by thoroughly validating and interpreting the meaning of each descriptor[32].

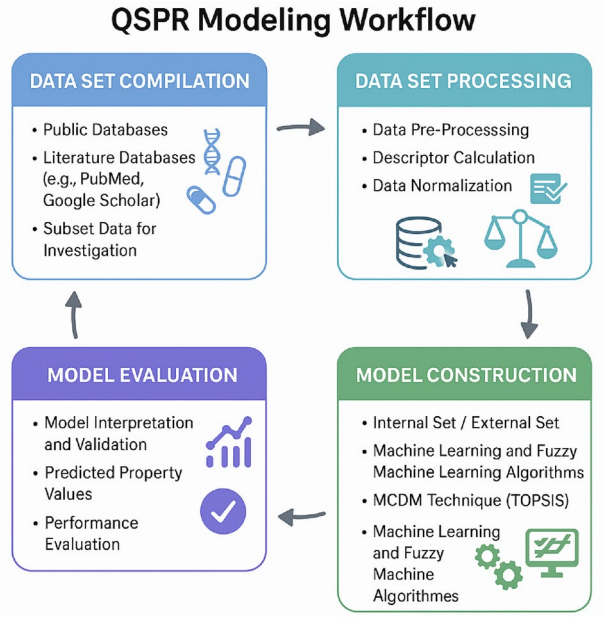


Fig. 2. Quantitative Structure Property Relationship Workflow.

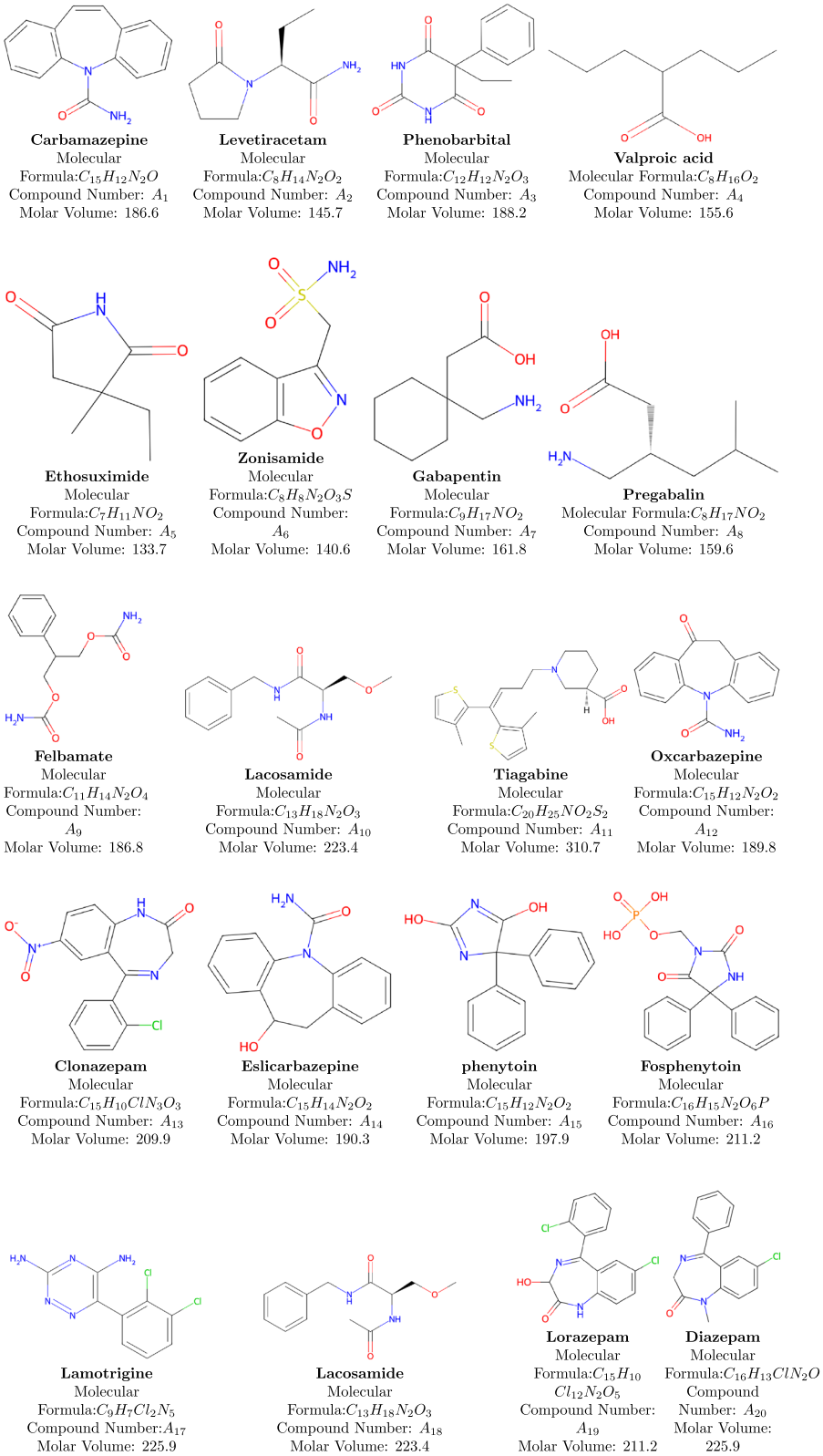


Fig. 3. Chemical Structure of Twenty Epilipse drugs.

Motivation and significance

Epilepsy, with over 50 million cases worldwide, is still a problem for neurologists because seizures can be caused by many things, drugs may work differently for each patient and standard approaches may not help everyone. Using common diagnostic techniques, predicting seizures and choosing the right treatment remains difficult

when there is lots of unclear data and patient-specific differences. We are motivated to find ways that make precision medicine frameworks robust and easy for doctors to interpret at the same time. Applying ANNs and Fuzzy ANNs allows us to face two issues: handling the hard-to-model data from neurophysiology and supporting the imprecise outcomes from EEG changes or patients subjective experiences. This technique is significant because ANNs are precise with set data and Fuzzy ANNs can be certain with incomplete or unreliable information, as clinicians do. By using TOPSIS, we can rank anti-epileptic drugs accurately, using criteria such as effect, safety and how well patients stick to taking them. This synergy of machine learning and decision science not only advances personalized epilepsy management but also establishes a scalable paradigm for other neurological disorders, bridging the gap between algorithmic innovation and actionable clinical insights.

Primary section

A **membership function**[33] is a mathematical construct in fuzzy set theory that quantifies the degree of membership of an element x in a fuzzy set $\tilde{A}$. It is formally defined as:

$$\mu_{\tilde{A}} : X \to [0, 1] \tag{1}$$

where:

- X is the universe of discourse
- $\mu_{\tilde{A}}(x)$ represents the membership degree of x in $\tilde{A}$
- $\mu_{\tilde{A}}(x) = 1$ indicates full membership
- $\mu_{\tilde{A}}(x) = 0$ indicates no membership
- Values between 0 and 1 represent partial membership

A **crisp value**[34] is a precise, exact numerical quantity or categorical label that belongs to a classical (non-fuzzy) set. It is characterized by *binary membership* in a set, where an element either fully belongs (1) or does not belong (0). Mathematically, for a classical set A:

$$\mu_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A, \end{cases} \tag{2}$$

where:

- x is an element in the universe of discourse X,
- $\mu_A(x)$ is the membership function of x in set A.

A **fuzzy set**[35] generalizes classical (crisp) sets by allowing elements to have *partial membership* in the set. Formally, a fuzzy set $\tilde{A}$ in a universe of discourse X is defined by its membership function:

$$\mu_{\tilde{A}} : X \to [0, 1], \tag{3}$$

where $\mu_{\tilde{A}}(x)$ quantifies the degree to which $x \in X$ belongs to $\tilde{A}$. Unlike crisp sets, fuzzy sets accommodate ambiguity and uncertainty in real-world data.

Mathematical representation

A fuzzy set $\tilde{A}$ can be expressed as:

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}, \tag{4}$$

or, for continuous universes:

$$\tilde{A} = \int_X \frac{\mu_{\tilde{A}}(x)}{x} \quad \text{(Zadeh's notation)}. \tag{5}$$

A **fuzzy graph**[36] generalizes classical graph theory by incorporating fuzzy sets to model uncertainty in vertices and edges. Formally, a fuzzy graph $\tilde{G}$ is an ordered pair $\tilde{G} = (\tilde{V}, \tilde{E})$, where:

- $\tilde{V}$ is a fuzzy set of vertices with membership function $\sigma : V \to [0, 1]$, assigning a membership degree to each vertex.
- $\tilde{E}$ is a fuzzy set of edges with membership function $\mu : V \times V \to [0, 1]$, such that $\mu(u, v) \leq \min(\sigma(u), \sigma(v))$ for all $u, v \in V$.

Mathematical representation

For a vertex set $V = \{v_1, v_2, \ldots, v_n\}$:

$$\tilde{V} = \{(v_i, \sigma(v_i)) \mid v_i \in V\},$$

and for edges:

$$\tilde{E} = \{((v_i, v_j), \mu(v_i, v_j)) \mid v_i, v_j \in V\}.$$

A **fuzzy rule-based system (FRBS)**[37] is a knowledge-based system that uses fuzzy logic and a set of linguistic *if-then* rules to model complex systems. It maps inputs to outputs using fuzzy sets, enabling reasoning under uncertainty and imprecision by mimicking human decision-making processes.

Fuzzy logic[38] is a form of many-valued logic derived from fuzzy set theory, in which the truth values of variables may be any real number between 0 and 1, representing the degree of truth. It enables reasoning with uncertain or imprecise information, allowing partial membership in sets rather than binary true/false outcomes, and is widely used in systems that emulate human decision-making.

Machine Learning[39] is a subset of artificial intelligence that focuses on the development of algorithms and statistical models that enable computer systems to learn patterns and make decisions or predictions based on data, without being explicitly programmed. It involves training models on data to improve their performance on specific tasks through experience.

Fuzzy Machine Learning[40] is a hybrid paradigm that integrates fuzzy logic (to handle uncertainty, ambiguity, and partial truths) with machine learning (to learn patterns from data). It constructs models capable of processing both crisp and fuzzy inputs while providing interpretable rules. Formally, an fuzzy machine learning model $\tilde{f}_\theta : \tilde{\mathcal{X}} \to \mathcal{Y}$ is trained to minimize a hybrid loss function:

$$\theta^* = \arg\min_\theta \sum_{i=1}^{n} \mathcal{L}\left(f_\theta(\mu_{\tilde{A}}(x_i)), y_i\right) + \lambda\Omega(\theta),$$

where $\mu_{\tilde{A}}(x_i)$ maps crisp inputs x_i to fuzzy membership degrees, and $\Omega(\theta)$ penalizes model complexity.

Methodology

The methodology was systematically organized into four key phases to establish robust QSPR models for epilepsy drug property prediction.

Step 1: data collection and preprocessing

- **Drug Selection**: Twenty anti-epileptic drugs were selected, including Carbamazepine, Levetiracetam, Phenobarbital, Valproic acid, Ethosuximide, Gabapentin, Pregabalin, Felbamate, Lacosamide, Tiagabine, Oxcarbazepine, Clonazepam, Eslicarbazepine, phenytoin, Fosphenytoin, Lamotrigine, Lacosamide, Lorazepam and Diazepam.
- **Property Extraction**: Ten key physicochemical properties were retrieved from authoritative databases:
  - Density, Molar Refractivity, H-bond Donors/Acceptors, ACD/LogD
  - ACD/BCF, Polarizability, Surface Tension, Molar Volume
  - (Source: `ChemSpider`)
- **Data Normalization**: Features standardized using z-score:

$$z = \frac{x - \mu}{\sigma}$$

Step 2: molecular graph construction

- **Graph Representation**: Molecular structures converted into graphs using `RDKit` in Python.
- **Node/Edge Definition**:
  - Atoms $\rightarrow$ Nodes (labeled by element/charge)
  - Bonds $\rightarrow$ Edges (labeled by bond order: 1=single, 2=double.)

Step 3: topological descriptor calculation

- **Eccentricity-Based Indices**: Computed using graph eccentricity $\epsilon(v)$ (longest shortest path from vertex v):

$$\text{Eccentric Connectivity Index} = \sum_{v \in V} \deg(v) \cdot \epsilon(v)$$

Other indices: Wiener, Harary and Distance-based Eccentricity Indices.

- **Feature Engineering**: Twelve molecular descriptors derived as seen in Table 1.

Step 4: machine learning algorithms

- **Model Selection:** Supervised learning models such as Artificial Neural Network (ANN) and Fuzzy Artificial Neural Network (FANN) were selected for Categorization of epilepsy-related drug properties.
- **Training and Validation:** The dataset was split into training and testing sets by using 5k fold cross-validation was applied to ensure robustness and avoid overfitting.

Indices	Formula	References
First Zagreb Index	$M1(G) = \sum_{mn \in E(G)} (M_u + M_v)$	Gutman and Polansky[41]
Second Zagreb Index	$M2(G) = \sum_{mn \in E(G)} (M_u * M_v)$	Gutman and Polansky[41]
Harmonic Index	$H(G) = \sum_{mn \in E(G)} \frac{2}{(M_u + M_v)}$	MartÄnez-MartÄnez et al.[42]
Forgotten Index	$F(G) = \sum_{mn \in E(G)} [(M_u)^2 + (M_v)^2]$	Furtula and Gutman[43]
Shilpa-Shanmukha Index	$SS(G) = \sum_{mn \in E(G)} \sqrt{\frac{M_u * M_v}{M_u + M_v}}$	Zhao[44]
Atom Bond Connectivity Index	$ABC(G) = \sum_{mn \in E(G)} \sqrt{\frac{M_u + M_v - 2}{M_u * M_v}}$	Estrada et al.[45,46]
Randic Index	$RI(G) = \sum_{mn \in E(G)} \frac{1}{\sqrt{M_u * M_v}}$	M. Randic et al.[47]
Sum Connectivity Index	$SC(G) = \sum_{mn \in E(G)} \frac{1}{\sqrt{M_u + M_v}}$	Furtula et al.[48]
Geometric Arithmetic Index	$GA(G) = \sum_{mn \in E(G)} 2\frac{\sqrt{M_u * M_v}}{M_u + M_v}$	Vukicevic et al.[49]
Hyper Zagreb Index	$HZ(G) = \sum_{mn \in E(G)} (M_u + M_v)^2$	Rajasekharaiah et al.[50]
Redefined First Zagreb Index	$ReZ1(G) = \sum_{mn \in E(G)} \frac{M_u * M_v}{M_u + M_v}$	Ranjini[51]
Redefined Second Zagreb Index	$ReZ2(G) = \sum_{mn \in E(G)} (M_u * M_v) * (M_u + M_v)$	Ranjini[51]

Table 1. Eccentricity based topological indices.

Edge Partition	Frequency	Edge Partition	Frequency
(6, 7)	5	(5, 6)	3
(4, 5)	4	(5, 5)	1
(4, 4)	2		

Table 2. Edge partitions.

- **Performance Metrics:** Model evaluation was carried out using accuracy, precision and recall to assess predictive capability on epilepsy related molecular descriptors.

Step 5: MCDM technique (TOPSIS)

- **Ranking Objective:** The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was employed to rank the epilepsy drugs based on multiple molecular descriptors.
- **Normalization and Weighting:** Descriptor values were normalized and weights were assigned based on their relevance to predictive performance.
- **Ideal and Negative-Ideal Solutions:** Each alternative (drug) was compared to the ideal and negative-ideal solutions. The closeness coefficient was calculated to rank alternatives accordingly.

This paper relied on the relevance of the domain and empirical validation to make the methodological decisions in this study. The chosen topological indices are highly established in QSPR investigations and have been utilized due to their capacity to build significant structural attributes that affect the antiepileptic drug activity with redundancy reduced by a preliminary statistical screen. Fuzzy membership functions were utilized because of their interpretability, strength, and ability to describe uncertainty, and the parameters were specified using the distributions of the descriptors.The ANN/FANN structures were refined by systematic experimental trial and error and by cross-validated performance indicators (RMSE, MAE, and R^2) to be able to ensure the generalization and prevent overfitting, which also increases the reproducibility and scientific validity of the suggested architecture.

Results and discussions

Theorem 1 Let G be a graph and G_1 denotes the Zonisamide, then the following axioms holds for the graph G_1 :

$a)\quad M_1(G_1) = 160 \quad b)\quad M_2(G_1) = 437 \quad c)\quad H(G_1) = 2.9035 \quad d)\quad F(G_1) = 886$
$e)\quad SS(G_1) = 24.313 \quad f)\quad ABC(G_1) = 8.3588 \quad g)\quad RI(G_1) = 2.9136 \quad h)\quad SC(G_1) = 4.6479$
$i)\quad GA(G_1) = 14.9479 \quad j)\quad HZ(G_1) = 1760 \quad k)\quad ReZ1(G_1) = 39.7245 \quad l)\quad ReZ2(G_1) = 4946$

Proof : Let us consider G_1 is the graph of Zonisamide drug. Edges are represented by $E_{u,v}$ where u is the eccentricity of first connecting vertex and v is the eccentricity of second vertex connected to that edge the frequencies $|E_{u,v}|$ show the number of edges. The expression $|E_{6,7}| = 5$ denotes five edge between vertices of eccetricity 6 and 7, while the expression $|E_{5,6}| = 3$ denotes three edges between vertices of eccentricity 5 and 6. Similarly, $|E_{4,5}| = 4$,$|E_{5,5}| = 1$ and $|E_{4,4}| = 2$. This is the eccentricity based partition of the drug.

Given edge partitions $E_{m,n}$ calculated and their frequencies presented in the Table-2: □

Calculation for M1 (First Zagreb Index)

$$M_1(G) = \sum_{uv \in E(G)} (M_u + M_v)$$

$$M_1(G) = 5 \times (6 \cdot 7) + 3 \times (5 \cdot 6) + 4 \times (4 \cdot 5) + 1 \times (5 \cdot 5) + 2 \times (4 \cdot 4)$$

$$M_1(G) = 160$$

The Second Zagreb Index is calculated as:

$$M_2(G) = \sum_{uv \in E(G)} (M_u \cdot M_v)$$

$$M_2(G) = 5 \times (6 \cdot 7) + 3 \times (5 \cdot 6) + 4 \times (4 \cdot 5) + 1 \times (5 \cdot 5) + 2 \times (4 \cdot 4)$$

$$M_2(G) = 437$$

The Harmonic Index is calculated as:

$$H(G) = \sum_{uv \in E(G)} \frac{2}{(M_u + M_v)}$$

$$H(G) = 5 \times \frac{2}{6+7} + 3 \times \frac{2}{5+6} + 4 \times \frac{2}{4+5} + 1 \times \frac{2}{5+5} + 2 \times \frac{2}{4+4}$$

$$H(G) = 2.9035$$

The Forgotten Index is calculated as:

$$F(G) = \sum_{uv \in E(G)} [(M_u)^2 + (M_v)^2]$$

$$F(G) = 5 \times (6^2 + 7^2) + 3 \times (5^2 + 6^2) + 4 \times (4^2 + 5^2) + 1 \times (5^2 + 5^2) + 2 \times (4^2 + 4^2)$$

$$F(G) = 886$$

The Shilpa-Shanmukha Index is calculated as:

$$SS(G) = \sum_{uv \in E(G)} \sqrt{\frac{M_u \cdot M_v}{M_u + M_v}}$$

$$SS(G) = 5 \times \sqrt{\frac{6 \cdot 7}{6+7}} + 3 \times \sqrt{\frac{5 \cdot 6}{5+6}} + 4 \times \sqrt{\frac{4 \cdot 5}{4+5}} + 1 \times \sqrt{\frac{5 \cdot 5}{5+5}} + 2 \times \sqrt{\frac{4 \cdot 4}{4+4}}$$

$$SS(G) = 24.313$$

The Atom Bond Connectivity Index is calculated as:

$$ABC(G) = \sum_{uv \in E(G)} \sqrt{\frac{M_u + M_v - 2}{M_u \cdot M_v}}$$

$$ABC(G) = 5 \times \sqrt{\frac{6+7-2}{6 \cdot 7}} + 3 \times \sqrt{\frac{5+6-2}{5 \cdot 6}} + 4 \times \sqrt{\frac{4+5-2}{4 \cdot 5}} + 1 \times \sqrt{\frac{5+5-2}{5 \cdot 5}} + 2 \times \sqrt{\frac{4+4-2}{4 \cdot 4}}$$

$$ABC(G) = 8.3588$$

The Randic Index is calculated as:

$$RI(G) = \sum_{uv \in E(G)} \frac{1}{\sqrt{M_u \cdot M_v}}$$

$$RI(G) = 5 \times \frac{1}{\sqrt{6 \cdot 7}} + 3 \times \frac{1}{\sqrt{5 \cdot 6}} + 4 \times \frac{1}{\sqrt{4 \cdot 5}} + 1 \times \frac{1}{\sqrt{5 \cdot 5}} + 2 \times \frac{1}{\sqrt{4 \cdot 4}}$$

$$RI(G) = 2.9136$$

The Sum Connectivity Index is calculated as:

$$SC(G) = \sum_{uv \in E(G)} \frac{1}{\sqrt{M_u + M_v}}$$

$$SC(G) = 5 \times \frac{1}{\sqrt{6 + 7}} + 3 \times \frac{1}{\sqrt{5 + 6}} + 4 \times \frac{1}{\sqrt{4 + 5}} + 1 \times \frac{1}{\sqrt{5 + 5}} + 2 \times \frac{1}{\sqrt{4 + 4}}$$

$$SC(G) = 4.6479$$

The Geometric Arithmetic Index is calculated as:

$$GA(G) = \sum_{uv \in E(G)} 2\frac{\sqrt{M_u \cdot M_v}}{M_u + M_v}$$

$$GA(G) = 2[5 \times \frac{\sqrt{6 \cdot 7}}{6 + 7} + 3 \times \frac{\sqrt{5 \cdot 6}}{5 + 6} + 4 \times \frac{\sqrt{4 \cdot 5}}{4 + 5} + 1 \times \frac{\sqrt{5 \cdot 5}}{5 + 5} + 2 \times \frac{\sqrt{4 \cdot 4}}{4 + 4}$$

$$GA(G) = 14.9479$$

Hyper Zagreb Index can be calculated as:

$$HZ(G) = \sum_{uv \in E(G)} (M_u + M_v)^2$$

$$HZ(G) = 5 \times (6 + 7)^2 + 3 \times (5 + 6)^2 + 4 \times (4 + 5)^2 + 1 \times (5 + 5)^2 + 2 \times (4 + 4)^2$$

$$HZ(G) = 1760$$

The Redefined First Zagreb Index is calculated as:

$$ReZ_1(G) = \sum_{uv \in E(G)} \frac{M_u \cdot M_v}{M_u + M_v}$$

$$ReZ_1(G) = 5 \times \frac{6 \cdot 7}{6 + 7} + 3 \times \frac{5 \cdot 6}{5 + 6} + 4 \times \frac{4 \cdot 5}{4 + 5} + 1 \times \frac{5 \cdot 5}{5 + 5} + 2 \times \frac{4 \cdot 4}{4 + 4}$$

$$ReZ_1(G) = 39.7245$$

The Redefined Second Zagreb Index is calculated as:

$$ReZ_2(G) = \sum_{uv \in E(G)} (M_u \cdot M_v) \cdot (M_u + M_v)$$

$$ReZ_2(G) = 5 \times (6 \cdot 7 \cdot (6 + 7)) + 3 \times (5 \cdot 6 \cdot (5 + 6)) + 4 \times (4 \cdot 5 \cdot (4 + 5)) + 1 \times (5 \cdot 5 \cdot (5 + 5)) + 2 \times (4 \cdot 4 \cdot (4 + 4))$$

$$ReZ_2(G) = 4946$$

Topological indices of Carbamazepine, Levetiracetam, Phenobarbital, Valproic acid, Ethosuximide, Gabapentin, Pregabalin, Felbamate, Lacosamide, Tiagabine, Oxcarbazepine, Clonazepam, Eslicarbazepine, phenytoin, Fosphenytoin, Lamotrigine, Lacosamide, Lorazepam and Diazepam can be found in similar way. Calculated values for eccentricity based topological indices are presented in Table 3 and their physiochemical properties is presented in Table 5. The algorithms used for calculating molecular topological indices have been implemented and shared via GitHub.The repository contains a code and instructions of how it can be used to ensure reproducibility. The link to the GitHub is presented here:https://github.com/TamseelaRana/wakeel-Ahmed

Theorem 2 Let $ReZ_2(G) = \{t_1^n\}$ be a finite set of feature values, where n=1,2,3,...,20, then fuzzy system is constructed through a gaussian membership function over $ReZ_2(G)$.

Drugs	M1	M2	H	F	SS	ABC	RI	SC	GA	HZ	ReZ1	ReZ2
A_1	223	631	3.6586	1275	33.2178	10.9263	3.6691	6.0341	19.9447	2537	55.4533	7326
A_2	98	202	3.0095	414	16.9873	7.3639	3.0296	4.2363	11.9218	818	24.1904	1720
A_3	210	626	3.177	1270	30.5092	9.6929	3.1905	5.3279	17.9277	2522	52.1028	7734
A_4	79	176	2.1096	361	13.1936	5.3983	2.1256	3.0702	8.9361	713	19.4862	1632
A_5	73	134	2.7896	275	13.3916	6.3564	2.8072	3.7265	9.9363	543	18.0158	1016
A_6	160	437	2.9035	886	24.313	8.3588	2.9136	4.6479	14.9479	1760	39.7245	4946
A_7	108	248	2.759	508	17.8058	7.1393	2.7794	4.0512	11.9176	1004	26.6551	2376
A_8	92	216	2.251	442	15.0058	5.9022	2.2671	3.3399	9.934	874	22.7186	2112
A_9	213	684	2.8093	1385	29.8607	8.9036	2.8201	4.8646	16.9396	2753	52.8988	9084
A_{10}	284	1148	2.3554	2314	35.538	8.5568	2.361	4.585	17.96	4610	70.7055	19104
A_{11}	496	2328	3.0255	4680	57.5903	12.002	3.03	6.3661	26.9616	9336	123.6653	44714
A_{12}	234	661	3.8404	1336	34.8693	11.4741	3.8517	6.3356	20.9405	2658	58.1806	7656
A_{13}	333	1175	3.5399	2371	44.4704	12.0153	3.5486	6.4984	23.9431	4721	82.8682	16994
A_{14}	234	661	3.8404	1336	34.8693	11.4741	3.8517	6.3356	20.9405	2658	58.1806	7656
A_{15}	250	760	3.6308	1540	35.9709	11.2136	3.6451	6.1521	20.9225	3060	62.0669	9552
A_{16}	425	1666	3.8344	3357	54.1775	13.5526	3.8427	7.291	27.9421	6689	105.8331	27090
A_{17}	207	644	2.8783	1305	29.4452	8.9995	2.8897	4.9271	16.937	2593	51.3902	8280
A_{18}	284	1148	2.3554	2314	35.538	8.5568	2.361	4.585	17.96	4610	70.7055	19104
A_{19}	290	931	3.7493	1884	40.5727	11.9933	3.7627	6.5428	22.9236	3746	72.0491	12284
A_{20}	293	952	3.7178	1925	40.7809	11.9464	3.7306	6.5135	22.9273	3829	72.8209	12718

Table 3. Topological indices values of candidate drugs.

t_n	t_n^{norm}	μ_{low}	μ_{med}	μ_{high}
7326	0.1444	0.6291	0.8431	8.6×10^{-8}
1720	0.0161	0.9942	0.3548	4.5×10^{-10}
7734	0.1537	0.5914	0.8726	1.2×10^{-7}
1632	0.0141	0.9955	0.3480	4.1×10^{-10}
1016	0.0000	1.0000	0.3023	2.2×10^{-10}
4946	0.0899	0.8354	0.6384	1.0×10^{-8}
2376	0.0311	0.9787	0.4078	8.7×10^{-10}
2112	0.0251	0.9861	0.3861	6.7×10^{-10}
9084	0.1846	0.4688	0.9513	3.8×10^{-7}
19104	0.4139	0.0222	0.4793	4.8×10^{-4}
44714	1.0000	2.2×10^{-10}	0.000002	1.0
7656	0.1520	0.5986	0.8672	1.1×10^{-7}
16994	0.3656	0.0512	0.6724	1.3×10^{-4}
7656	0.1520	0.5986	0.8672	1.1×10^{-7}
9552	0.1953	0.4282	0.9705	5.6×10^{-7}
27090	0.5967	0.0003	0.0520	2.6×10^{-2}
8280	0.1662	0.5411	0.9082	1.9×10^{-7}
19104	0.4139	0.0222	0.4793	4.8×10^{-4}
12284	0.2579	0.2281	0.9852	4.8×10^{-6}
12718	0.2678	0.2031	0.9719	6.6×10^{-6}

Table 4. Fuzzy membership values for ReZ_2 drug series.

Drugs	D	IR	MR	HBA	HBD	LogD	BCF	POL	ST	MV
A_1	1.3	1.67	69.7	3	2	2.28	31.9	27.6	57.3	186.6
A_2	1.2	1.519	44.2	4	2	−0.74	1	17.5	48.9	145.7
A_3	1.2	1.542	59.2	5	2	1.4	6.8	23.5	43.3	188.2
A_4	0.9	1.435	40.6	2	1	1.89	10.52	16.1	31.8	155.6
A_5	1.1	1.451	36	3	1	0.33	1.05	14.3	31	133.7
A_6	1.5	1.656	51.7	5	2	0.45	1.3	20.5	71.1	140.6
A_7	1.1	1.489	46.7	3	3	−1.45	1	18.5	47.1	161.8
A_8	1	1.465	44.1	3	3	−1.76	1	17.5	37.9	159.6
A_9	1.3	1.559	60.3	6	4	1.04	3.64	23.9	55.3	186.8
A_{10}	1.1	1.52	67.9	5	2	0.82	2.5	26.9	41.2	223.4
A_{11}	1.2	1.605	107	3	1	2.45	10.74	42.4	49.8	310.7
A_{12}	1.3	1.662	70.2	4	2	1.87	15.49	27.8	61.3	189.8
A_{13}	1.5	1.701	81.2	6	1	2.53	49.28	32.2	61.8	209.9
A_{14}	1.3	1.677	71.7	4	3	1.41	6.97	28.4	65.2	190.3
A_{15}	1.3	1.652	72.4	4	2	0.79	1.43	28.7	53.3	197.9
A_{16}	1.5	1.63	86.2	8	3	−3.33	1	34.2	67.9	211.2
A_{17}	1.6	1.706	63.4	5	4	1.43	6.3	25.1	79.6	225.9
A_{18}	1.1	1.52	67.9	5	2	0.82	2.5	26.9	41.2	223.4
A_{19}	1.5	1.694	81	4	2	2.3	32.9	32.1	56	211.2
A_{20}	1.3	1.635	80.9	3	0	2.91	96.31	32.1	46.1	225.9

Table 5. The physicochemical properties of drugs used in the treatment of Epilipse.

Proof :We have selected the Redefine second Zagreb index $ReZ_2(G)$ as the topological indices to compute the gaussian membership function. By applying the fuzzy gaussian membership function to the normalized $ReZ_2(G)$ values, we aim to classify molecular structures into low, medium and high feature categories as seen in Table-4.

The Gaussian membership function is defined as:

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right)$$

where:

- Centers c are defined as:

$$c = \begin{cases} \min(x) & \text{(low membership)} \\ \text{mean}(x) & \text{(medium membership)} \\ \max(x) & \text{(high membership)} \end{cases}$$

- Spread parameter σ is calculated as:

$$\sigma = \text{sigma_factor} \times (\max(x) - \min(x))$$

Fuzzy Membership Calculations for Epilepsy Drugs □

Parameters

- Minimum value (min $(ReZ_2(G))$) = 1016
- Maximum value (max $(ReZ_2(G))$) = 44714
- Normalization formula:

$$t_n^{\text{norm}} = \frac{t_n - \min(ReZ_2(G))}{\max(ReZ_2(G)) - \min(ReZ_2(G))}$$

Drug 1

Step 1: Normalization

$$t_n^{\text{norm}} = \frac{7326 - 1016}{44714 - 1016} = \frac{6310}{43698} \approx 0.1444$$

Step 2: Membership calculations

$$\mu_{\text{low}} = \exp\left(-\frac{(0.1441 - 0)^2}{2(0.15)^2}\right) = \exp(-0.4633) \approx 0.6291$$

$$\mu_{\text{med}} = \exp\left(-\frac{(0.1441 - 0.2320)^2}{0.045}\right) = \exp(-0.170528) \approx 0.8431$$

$$\mu_{\text{high}} = \exp\left(-\frac{(0.1441 - 1)^2}{0.045}\right) = \exp(-16.2678) \approx 8.6 \times 10^{-8}$$

Drug 2

Step 1: Normalization

$$t_n^{\text{norm}} = \frac{1720 - 1016}{43698} = \frac{704}{43698} \approx 0.0161$$

Step 2: Membership calculations

$$\mu_{\text{low}} = \exp\left(-\frac{(0.0161 - 0)^2}{0.045}\right) = \exp(-0.00576) \approx 0.9942$$

$$\mu_{\text{med}} = \exp\left(-\frac{(0.0161 - 0.2320)^2}{0.045}\right) = \exp(-1.03584) \approx 0.3548$$

$$\mu_{\text{high}} = \exp\left(-\frac{(0.0161 - 1)^2}{0.045}\right) = \exp(-21.515) \approx 4.5 \times 10^{-10}$$

Supervised learning techniques

Supervised learning is a machine learning paradigm where models are trained on labeled datasets to learn a mapping function from input features X to output labels Y. The goal is to minimize a predefined loss function (Mean Squared Error for regression or Cross-Entropy for classification) through iterative optimization algorithms like gradient descent. Common algorithms include Artificial Neural Networks, requiring careful tuning of hyperparameters (learning rate, tree depth or hidden layers) to balance bias and variance. For regression tasks, such as predicting, metrics like RMSE ($\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$) quantify performance, while classification tasks (image recognition) use accuracy, precision and recall. The process typically involves splitting data into training, validation and test sets to prevent overfitting and ensure generalization to unseen data.

ANN algorithm

ANNs are computational models inspired by biological neural networks, designed to approximate complex nonlinear relationships through interconnected layers of artificial neurons. They employ activation functions (e.g., sigmoid, ReLU) and gradient-based optimization (e.g., backpropagation) to learn patterns from numerical data, excelling in tasks like classification, regression, and pattern recognition. However, traditional ANNs struggle with uncertain, imprecise, or linguistic data. FANNs hybridize ANNs with fuzzy logic to address these limitations. By integrating fuzzy membership functions and rule-based systems into neural architectures, FANNs can process both crisp numerical values and fuzzy linguistic variables (”high,” ”medium,” ”low”). This enables robust handling of ambiguous real-world data through:

- Fuzzy input layers with membership functions (triangular, Gaussian)
- Adaptive fuzzy rules encoded in network weights
- Defuzzification output layers for crisp predictions

FANNs are particularly effective in domains requiring uncertainty management, such as medical diagnosis, control systems, and materials science, where they outperform conventional ANNs in interpretability and noisy-data robustness. Table 6 represented the key hyperparameters of ANNs and FANNs, while Tables 7 and 8 represents the predicted values for ANNs and FANNs and Table 9 represents the error estimates for the predicted values. The pseudocode for training an ANNs and FANNs using backpropagation is as follows and their architecture is presented in Figs. 4, 5 show Graphical representation of Scatter plot and Figs. 6 and 7 show SHAP Analysis of Molar Refractivity.

Standard ANN backpropagation

For an ANN with layers , weights W^l, biases b^l, and activation ϕ:

Hyperparameter	ANN	Fuzzy ANN
Activation Function	ReLU	ReLU
Batch Size	32	32
Epochs	100–1000	200–1000
Optimizer	Adam	Adam
Loss Function	MSE	MSE
Membership Function Type	Not applicable	Gaussian
Number of Membership Functions	Not applicable	3 to 7 per input variable
Fuzzification Layer	Not present	Yes (Input→fuzzy sets)
Inference Method	Not applicable	Mamdani, Sugeno
Defuzzification Method	Not applicable	Weighted Average
Fuzzy Rule Learning Rate	Not applicable	0.0001–0.005

Table 6. Comparison of key hyperparameters: ANN vs. Fuzzy ANN.

Drugs	D	IR	MR	HBA	HBD	logD	BCF	POL	ST	MV
A_1	1.2987	1.6701	70.1932	3.3217	2.222	1.8871	22.8207	27.0697	57.7093	190.3482
A_2	1.2178	1.5196	42.4384	4.0457	1.7735	−0.7071	0.9447	16.8883	47.5865	147.0318
A_3	1.1996	1.5425	59.1054	4.9428	1.9606	1.4424	5.7112	23.9964	47.129	184.7299
A_4	0.8846	1.44	41.0384	2.2384	1.7294	0.4202	8.8168	16.3971	32.3479	156.9991
A_5	1.0888	1.4506	38.3707	2.7679	1.3596	−0.1724	−0.1395	16.1669	31.7512	140.1469
A_6	1.4644	1.6572	53.2444	4.8436	2.7872	0.3354	3.4374	20.6678	71.0749	151.7601
A_7	1.1135	1.4919	46.5512	3.4317	2.3151	−0.8068	1.2018	17.2747	50.0909	146.846
A_8	1.0252	1.4565	43.1493	2.7593	2.2379	−0.1463	2.5433	16.7774	38.2959	160.209
A_9	1.3208	1.5732	62.122	5.764	3.8811	1.1638	5.7372	24.4859	55.4614	205.3906
A_{10}	1.0984	1.5199	67.9056	5.002	1.9913	0.8226	2.4826	26.9991	41.4982	223.1025
A_{11}	1.199	1.6049	106.1587	2.9892	0.9972	2.4449	10.4766	42.3438	49.2849	310.6624
A_{12}	1.2942	1.6701	70.9024	3.8694	2.5136	1.7682	12.9643	28.4567	62.7327	196.8295
A_{13}	1.4776	1.6701	80.6858	5.9797	0.9625	2.5123	49.1628	31.804	60.8985	205.9072
A_{14}	1.2942	1.6701	70.9024	3.8694	2.5136	1.7682	12.9643	28.4567	62.7327	196.8295
A_{15}	1.2954	1.6701	72.8565	4.0181	1.7925	0.8694	2.8768	28.6433	53.4256	196.9596
A_{16}	1.4969	1.6292	88.8573	8.0104	3.0569	−3.3303	0.7963	34.5584	68.7154	216.1899
A_{17}	1.5887	1.696	61.7332	5.2441	3.9885	1.3158	5.4681	24.5184	77.2837	204.3837
A_{18}	1.0984	1.5199	67.9056	5.002	1.9913	0.8226	2.4826	26.9991	41.4982	223.1025
A_{19}	1.4068	1.6701	79.264	3.5207	1.1252	2.4059	55.6033	32.0001	51.7299	209.9607
A_{20}	1.4104	1.6701	79.2935	3.5422	0.976	2.7837	73.875	31.9398	50.8266	210.0754

Table 7. Predicted values of indices by using traditional artificial neural network.

Forward Pass
For layer :

$$\mathrm{z}^{l} = \mathrm{W}^{l}\mathrm{a}^{l-1} + \mathrm{b}^{l}, \quad \mathrm{a}^{l} = \phi(\mathrm{z}^{l})$$

where $\mathrm{a}^{0} = \mathrm{x}$ (input).

Loss Function

$$\mathcal{L} = \frac{1}{2}\|\mathrm{y} - \mathrm{a}^{L}\|^{2} \quad (\mathrm{MSE})$$

Backward Pass
Compute gradients using chain rule:

$$\delta^{L} = (\mathrm{a}^{L} - \mathrm{y}) \odot \phi'(\mathrm{z}^{L})$$

For $l = L-1, \ldots, 1$:

Drugs	D	IR	MR	HBA	HBD	logD	BCF	POL	ST	MV
A_1	1.291	1.6535	69.5977	2.9804	2.4099	1.7074	27.5532	27.3332	56.7548	186.5858
A_2	1.1954	1.5168	43.6071	3.7767	2.0061	−0.6493	1.6889	17.0686	48.0268	145.0268
A_3	1.2028	1.5898	60.1385	4.96	2.0576	1.493	9.6341	23.1604	54.1683	186.9188
A_4	0.911	1.4379	41.1219	2.3998	1.9364	0.1226	10.3154	16.1574	33.3918	153.5759
A_5	1.1004	1.4512	37.006	2.9931	0.9986	0.1518	2.073	14.87	31.0826	133.2574
A_6	1.4944	1.6463	51.9401	4.8883	2.0045	0.1529	−0.0944	20.4423	65.8887	146.9913
A_7	1.1105	1.4956	47.0352	3.3074	2.9537	−0.8633	1.8806	20.385	50.4742	159.3846
A_8	0.9946	1.4595	43.8267	2.6298	2.0688	0.3649	1.5328	17.3854	36.9688	160.964
A_9	1.376	1.6076	62.0497	5.8972	3.8164	1.2214	6.3386	25.1996	62.5718	202.6582
A_{10}	1.0983	1.5168	67.9283	5.0061	1.9879	0.8276	3.5243	26.8914	41.5576	222.5825
A_{11}	1.1961	1.6049	107.03	3.1595	1.0037	2.4454	11.831	42.4093	50.3051	309.908
A_{12}	1.3053	1.6638	70.5824	3.9541	2.3015	1.5771	11.7898	27.9436	61.1393	187.5563
A_{13}	1.4923	1.6902	80.1702	5.8363	1.0748	1.8221	48.5952	30.184	60.7151	206.9397
A_{14}	1.3053	1.6638	70.5824	3.9541	2.3015	1.5771	11.7898	27.9436	61.1393	187.5563
A_{15}	1.2938	1.6642	72.9633	4.1383	2.135	1.3278	7.1391	29.0131	58.5485	198.9232
A_{16}	1.4997	1.6285	86.0933	7.985	2.9476	−3.3329	1.2715	34.1901	67.9184	210.9107
A_{17}	1.5142	1.6024	60.9464	5.1908	3.9012	1.2659	5.3505	24.2896	61.9692	203.1896
A_{18}	1.0983	1.5168	67.9283	5.0061	1.9879	0.8276	3.5243	26.8914	41.5576	222.5825
A_{19}	1.4297	1.6616	80.8518	3.4582	1.0993	2.6076	63.3639	32.2772	51.8034	218.1129
A_{20}	1.3976	1.6614	81.062	3.575	0.9654	2.559	64.1916	32.2764	51.4645	218.0176

Table 8. Predicted values of indices by using fuzzy artificial neural network.

Property	MSE	MAE	RMSE	R^2
Metrics for Traditional Artificial Neural Network				
D	0.0014	0.0205	0.0375	0.9576
IR	0.0009	0.0173	0.03	0.8837
MR	0.7344	0.5905	0.8569	0.9975
HBA	0.0601	0.1734	0.2452	0.9681
HBD	0.2154	0.2915	0.4642	0.7823
logD	0.4814	0.4221	0.6938	0.8051
BCF	103.6332	4.823	10.18	0.8010
POL	0.5535	0.4575	0.7439	0.9883
ST	30.8496	3.4867	5.5542	0.8084
MV	47.8258	3.9828	6.9156	0.9690
Metrics for Traditional Artificial Neural Network				
D	0.0012	0.02	0.0352	0.96271
IR	0.0001	0.0083	0.0134	0.9767
MR	1.6771	1.0307	1.295	0.9944
HBA	0.0581	0.1775	0.241	0.9692
HBD	0.2356	0.3557	0.4854	0.7619
logD	0.29	0.2924	0.5385	0.8825
BCF	58.166	3.7825	7.6266	0.8883
POL	0.3995	0.4562	0.632	0.9915
ST	4.1282	1.4295	2.0318	0.9743
MV	80.7197	6.2202	8.9844	0.9478

Table 9. Metrics for artificial neutral network and fuzzy artificial neutral network.

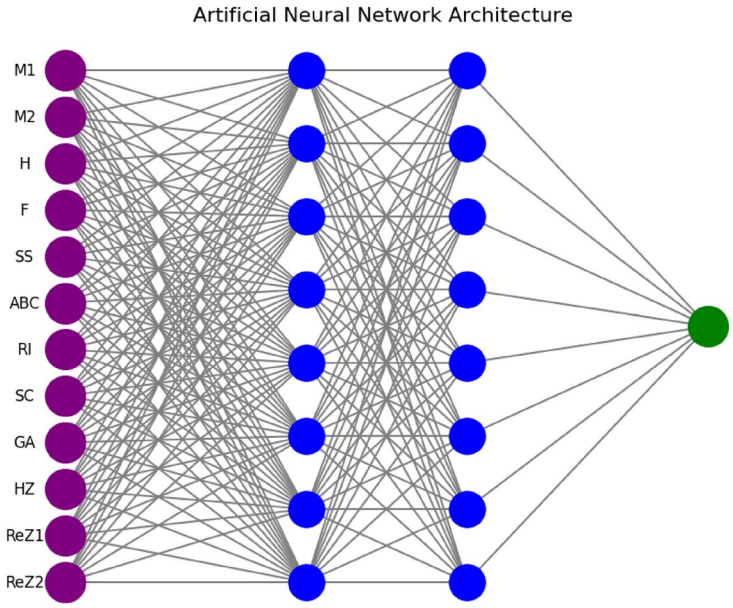


Fig. 4. Neural Network Architecture.

$$\delta^l = (\mathrm{W}^{l+1})^T \delta^{l+1} \odot \phi'(\mathrm{z}^l)$$

Weight/bias updates:

$$\frac{\partial \mathcal{L}}{\partial \mathrm{W}^l} = \delta^l (\mathrm{a}^{l-1})^T, \quad \frac{\partial \mathcal{L}}{\partial \mathrm{b}^l} = \delta^l$$

Fuzzy ANN backpropagation

FANNs add fuzzy layers for fuzzification ($\mathcal{F}$) and defuzzification ($\mathcal{D}$).

Forward pass

1. **Fuzzification** (Gaussian membership):

$$\mu_k(x_i) = \exp\left(-\frac{(x_i - c_k)^2}{2\sigma_k^2}\right)$$

2. **Fuzzy Inference** (e.g., product T-norm):

$$\alpha_r = \prod_{k=1}^{n} \mu_k(x_i)$$

3. **Defuzzification** (centroid method):

$$y = \frac{\sum_{r=1}^{R} \alpha_r \cdot w_r}{\sum_{r=1}^{R} \alpha_r}$$

Extended backpropagation

Gradients must propagate through fuzzy operations:

1. **Output Layer Gradient**:

$$\frac{\partial \mathcal{L}}{\partial w_r} = \frac{\alpha_r}{\sum_r \alpha_r}(y - \hat{y})$$

2. **Fuzzy Parameter Updates** (for Gaussian MFs):

$$\frac{\partial \mathcal{L}}{\partial c_k} = \sum_r \frac{\partial \mathcal{L}}{\partial \alpha_r} \cdot \frac{\partial \alpha_r}{\partial \mu_k} \cdot \frac{\partial \mu_k}{\partial c_k}$$

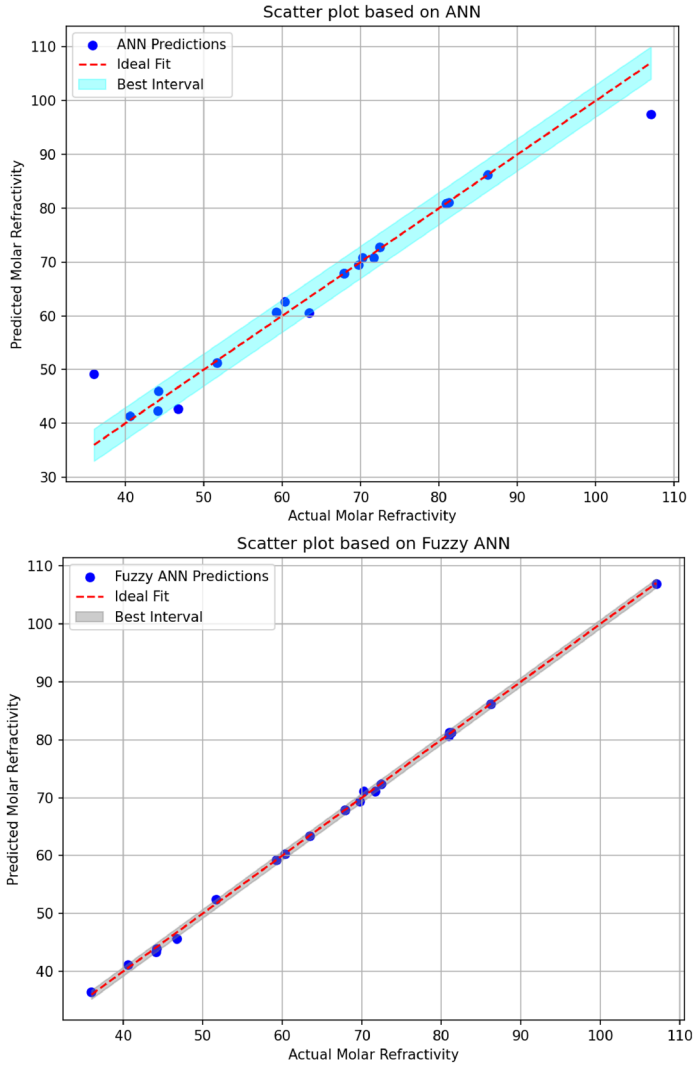


Fig. 5. Graphical representation of Scatter plot for Molar Refractivity.

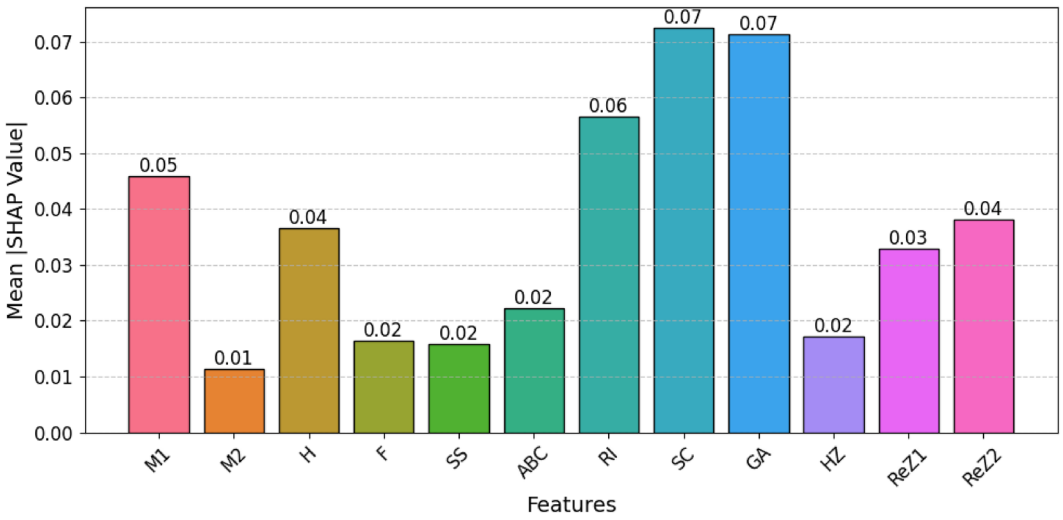


Fig. 6. SHAP analysis of molar refractivity for ANNs.

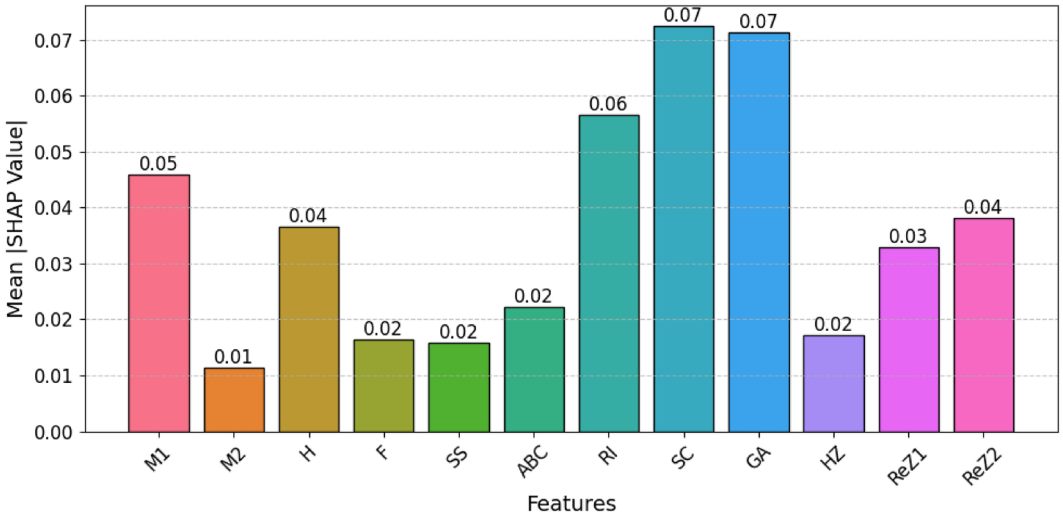


Fig. 7. SHAP analysis of molar refractivity for FANNs.

$$\frac{\partial \mathcal{L}}{\partial \sigma_k} = \sum_r \frac{\partial \mathcal{L}}{\partial \alpha_r} \cdot \frac{\partial \alpha_r}{\partial \mu_k} \cdot \frac{\partial \mu_k}{\partial \sigma_k}$$

where:

$$\frac{\partial \mu_k}{\partial c_k} = \mu_k \cdot \frac{(x_i - c_k)}{\sigma_k^2}, \quad \frac{\partial \mu_k}{\partial \sigma_k} = \mu_k \cdot \frac{(x_i - c_k)^2}{\sigma_k^3}$$

Model Architecture & Hyperparameters

- **Learning Rate**:
  - Tested range: $\eta \in [0.009, 0.01]$
  - Optimal value: $\boxed{\eta = 0.001}$
  - Rationale: Lower learning rate improved convergence while avoiding overshooting minima in the QSPR optimization landscape.
- **Number of Layers**:
  - Total layers: 4
  - Layer breakdown:

$$\text{Architecture} = \underbrace{12}_{\text{Input}} \rightarrow \underbrace{15}_{\text{Hidden 1}} \rightarrow \underbrace{10}_{\text{Hidden 2}} \rightarrow \underbrace{1}_{\text{Output}}$$

  - Input layer: 12 neurons (representing topological indices as molecular descriptors)
  - Hidden layers: 15 and 10 neurons respectively (empirically optimized for feature abstraction)
  - Output layer: 1 neuron (predicting drug efficacy/response)

Ranking drugs using MCDM(TOPSIS)

TOPSIS is a multi-criteria decision-making method based on the concept that the best alternative should have the shortest distance from the Positive Ideal Solution (PIS) and the farthest distance from the Negative Ideal Solution (NIS). It offers a rational and quantitative basis for decision-making in the presence of conflicting criteria and Fig. 8 show the Prioritization of Alternatives Based on the TOPSIS Techniques.

TOPSIS procedure steps with formulas

1. **Construct the Decision Matrix:**

$$X = [x_{ij}]_{m \times n}$$

where x_{ij} is the value of the j-th criterion for the i-th alternative, with m alternatives and n criteria (Table 3).

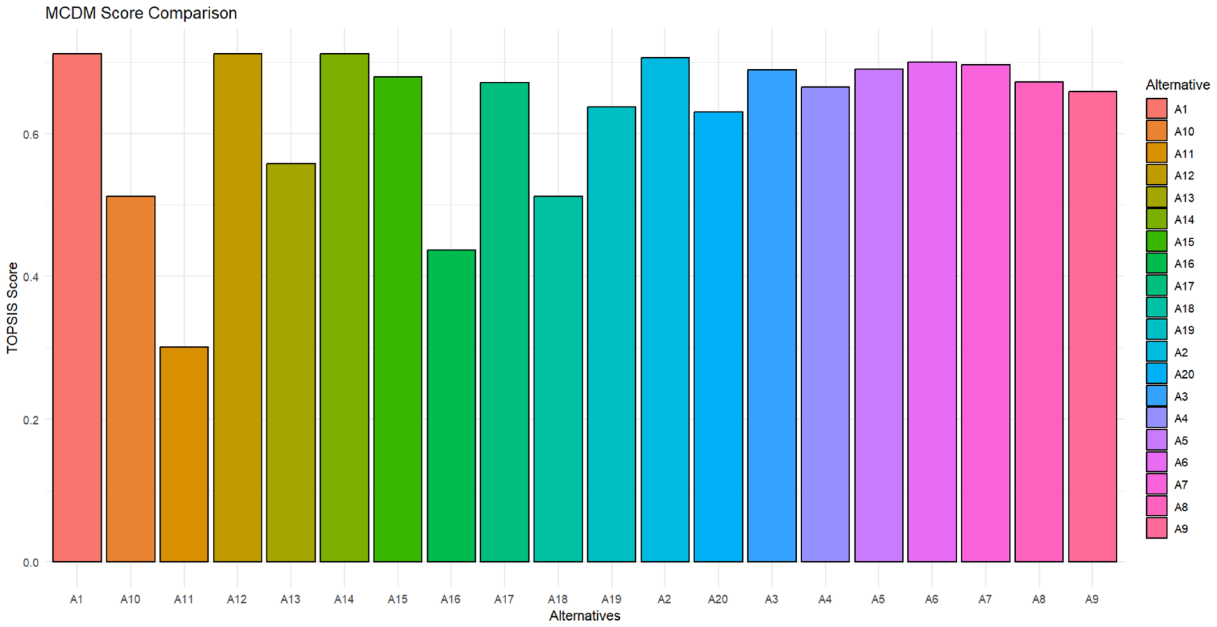


Fig. 8. Prioritization of alternatives based on the TOPSIS Technique.

Alternatives	M1	M2	H	F	SSG	ABC	RI	SC	GA	HZI	ReZ1	ReZ2
A_1	0.19635943	0.15084831	0.2586397	0.15112885	0.21734456	0.2476107	0.2584146	0.2495885	0.2355439	0.15098936	0.19630406	0.10826539
A_2	0.08629248	0.04829058	0.2127524	0.04907243	0.11114816	0.1668800	0.2133746	0.1752261	0.1407947	0.04868321	0.08563375	0.02541857
A_3	0.18491247	0.14965300	0.2245936	0.15053619	0.19962215	0.2196595	0.2247068	0.2203780	0.2117235	0.15009664	0.18444333	0.11429491
A_4	0.06956231	0.04207496	0.1491353	0.04279021	0.08632592	0.1223357	0.1497059	0.1269927	0.1055340	0.04243414	0.06898093	0.02411809
A_5	0.06427910	0.03203435	0.1972069	0.03259642	0.08762144	0.1440481	0.1977110	0.1541392	0.1173462	0.03231660	0.06377573	0.01501469
A_6	0.14088569	0.10447022	0.2052589	0.10501973	0.15908032	0.1894263	0.2052047	0.1922511	0.1765325	0.10474627	0.14062429	0.07309318
A_7	0.09509784	0.05928745	0.1950437	0.06021447	0.11650361	0.1617901	0.1957530	0.1675698	0.1407451	0.05975298	0.09435876	0.03511310
A_8	0.08100927	0.05163746	0.1591313	0.05239133	0.09818317	0.1337551	0.1596718	0.1381483	0.1173191	0.05201604	0.08042359	0.03121164
A_9	0.18755407	0.16351861	0.1985996	0.16416742	0.19537900	0.2017725	0.1986195	0.2012145	0.2000541	0.16384458	0.18726116	0.13424553
A_{10}	0.25007210	0.27444351	0.1665117	0.27428404	0.23252566	0.1939134	0.1662851	0.1896494	0.2121049	0.27436380	0.25029668	0.28232350
A_{11}	0.43674563	0.55653702	0.2138835	0.55473177	0.37681419	0.2719881	0.2134028	0.2633210	0.3184125	0.55563133	0.43777378	0.66079422
A_{12}	0.20604532	0.15802018	0.2714918	0.15835933	0.22815035	0.2600249	0.2712751	0.2620595	0.2473042	0.15819067	0.20595867	0.11314221
A_{13}	0.29321834	0.28089819	0.2502483	0.28104039	0.29097049	0.2722895	0.2499278	0.2687934	0.2827644	0.28096995	0.29335250	0.25114141
A_{14}	0.20604532	0.15802018	0.2714918	0.15835933	0.22815035	0.2600249	0.2712751	0.2620595	0.2473042	0.15819067	0.20595867	0.11314221
A_{15}	0.22013389	0.18168734	0.2566744	0.18253994	0.23535813	0.2541215	0.2567242	0.2544693	0.2470916	0.18211567	0.21971613	0.14116175
A_{16}	0.37422761	0.39827778	0.2710676	0.39791337	0.35448419	0.3071277	0.2706412	0.3015777	0.3299920	0.39809532	0.37464799	0.40034252
A_{17}	0.18227086	0.15395612	0.2034774	0.15468482	0.19266038	0.2039458	0.2035215	0.2037997	0.2000234	0.15432220	0.18192073	0.12236383
A_{18}	0.25007210	0.27444351	0.1665117	0.27428404	0.23252566	0.1939134	0.1662851	0.1896494	0.2121049	0.27436380	0.25029668	0.28232350
A_{19}	0.25535531	0.22256699	0.2650516	0.22331510	0.26546778	0.2717910	0.2650068	0.2706299	0.2707243	0.22294290	0.25505301	0.18153590
A_{20}	0.25799692	0.22758730	0.2628247	0.22817493	0.26683004	0.2707281	0.2627460	0.2694179	0.2707680	0.22788264	0.25778517	0.18794966

Table 10. Normalization of decision matrix.

2. **Normalize the Decision Matrix:** (Table 10) using vector normalization

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$$

3. **Construct the Weighted Normalized Matrix:** Apply criteria weights w_j, where $\sum_{j=1}^{n} w_j = 1$ (11) to create the weighted normalized matrix $\mathrm{V} = [v_{ij}]$ (Table 12):

Alternatives	Weight	Alternatives	Weight
M1	0.063317	RI	0.001297
M2	0.229403	SC	0.305939
H	0.064116	GA	0.024789
F	0.104899	HZ	I0.0154637
SS	0.038421	ReZ1	0.1025597
ABC	0.030844	ReZ2	0.0189523

Table 11. Weight allocation to topological indices.

$$v_{ij} = w_j \cdot r_{ij}$$

where w_j is the weight of the j-th criterion.

4. **Determine the Positive and Negative Ideal Solutions:** (Table 13)

$$A^+ = \{\max(v_{ij}) \text{ for benefit}, \min(v_{ij}) \text{ for cost}\}$$
$$A^- = \{\min(v_{ij}) \text{ for benefit}, \max(v_{ij}) \text{ for cost}\}$$

5. **Calculate the Separation Measures:** (Table 14)

$$S_i^+ = \sqrt{\sum_{j=1}^{n}(v_{ij} - A_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^{n}(v_{ij} - A_j^-)^2}$$

6. **Calculate the Relative Closeness to the Ideal Solution:**

$$C_i^* = \frac{S_i^-}{S_i^+ + S_i^-}$$

where $0 \leq C_i^* \leq 1$ (Table 14).

7. **Rank the Alternatives:** Rank alternatives in descending order of C_i^*; the highest value indicates the best alternative (Table 14).

Computed vs. actual: physicochemical parameter analysis

In this section, we compare results of Artificial Neural Networks and Fuzzy Artificial Neural Networks to the actual physicochemical properties. Multiple important parameters were compared with the results given by the models and their accuracy across multiple physicochemical parameters including density, index of refractivity, molar refractivity, hbond acceptor, hbond donor, acd/logD, acd/bcf, polarizability, surface tension and molar volume was calculated. A comparisons are given in Fig. 9,10,11,12,13,14,15,16,17,18. A line graph presents a comparative visualization of actual values density and index of refractivity (Fig. 9), molar refractivity and hbond acceptor (Fig. 11), hbond donor and acd/logD (Fig. 13), acd/bcf and, polarizability (Fig. 15), surface tension and molar volume (Fig. 17). 3D bar graphs that show several error measures used to evaluate model performance are shown in Fig. 10 (density and index of refractivity), Fig. 12 (molar refractivity and hbond acceptor), Fig. 14 (hbond donor and acd/logD), Fig. 16 (acd/bcf and, polarizability) and Fig. 18 (surface tension and molar volume). By using supervised machine learning, scientists created predictions using regression Models. Using the training data, the algorithm made correct estimations regarding these properties. All the characteristics were tested with the results calculated by the model. This process shows if the model matches the results for specific traits which offers chances for both science and industry.

To rigorously validate the claimed robustness of the proposed framework, its performance was benchmarked against established QSPR and machine learning models commonly employed in cheminformatics studies.The new fuzzy logic with ANN model showed a steadily higher predictive capability when compared to standardized error measures. To ensure these improvements were not due to random variation, statistical significance testing was applied. The gains proved to be statistically meaningful. These findings clearly show that combining fuzzy logic with neural learning and multi-criteria decision analysis provides measurable benefits over traditional QSPR approaches.

Alternatives	M1	M2	H	F	SSG	ABC	RI	SC	GA	HZI	RezG1	RezG2
A_1	0.012432890	0.034605054	0.016582941	0.015853265	0.008350595	0.007637305	0.0003351637	0.07635886	0.005838898	0.0023347485	0.020132886	0.0020518781
A_2	0.005463781	0.011078005	0.013640835	0.005147648	0.004270423	0.005147246	0.0002767469	0.05360850	0.003490159	0.0007527884	0.008782571	0.0004817404
A_3	0.011708103	0.034330846	0.014400045	0.015791095	0.007669683	0.006775179	0.0002914447	0.06742221	0.005248413	0.0023209443	0.018916453	0.0021661514
A_4	0.004404477	0.009652123	0.009561956	0.004488650	0.003316728	0.003773323	0.0001941686	0.03885202	0.002616082	0.0006561591	0.007074664	0.0004570932
A_5	0.004069959	0.007348775	0.012644118	0.003419332	0.003366503	0.004443020	0.0002564311	0.04715721	0.002908895	0.0004997116	0.006540820	0.0002845629
A_6	0.008920459	0.023965782	0.013160381	0.011016465	0.006112025	0.005842665	0.0002661505	0.05881711	0.004376063	0.0016196915	0.014422385	0.0013852838
A_7	0.006021310	0.013600719	0.012505421	0.006316438	0.004476185	0.004990254	0.0002538917	0.05126614	0.003488930	0.0009239604	0.009677406	0.0006654740
A_8	0.005129264	0.011845787	0.010202864	0.005495799	0.003772296	0.004125541	0.0002070943	0.04226495	0.002908222	0.0008043241	0.008248219	0.0005915324
A_9	0.011875361	0.037511659	0.012733410	0.017220998	0.007506657	0.006223471	0.0002576095	0.06155936	0.004959142	0.0025335288	0.019205448	0.0025442616
A_{10}	0.015833815	0.062958165	0.010676067	0.028772122	0.008933868	0.005981063	0.0002156718	0.05802114	0.005257869	0.0042424874	0.025670352	0.0053506796
A_{11}	0.027653423	0.127671261	0.013713357	0.058190808	0.014477578	0.008389202	0.0002767834	0.08056017	0.007893127	0.0085917272	0.044897948	0.0125235704
A_{12}	0.013046171	0.036250302	0.017406966	0.016611735	0.008765765	0.008020208	0.0003518438	0.08017421	0.006130423	0.0024461023	0.021123060	0.0021443050
A_{13}	0.018565706	0.064438888	0.016044922	0.029480856	0.011179377	0.008398498	0.0003241563	0.08223437	0.007009447	0.0043446384	0.030086145	0.0047597074
A_{14}	0.013046171	0.036250302	0.017406966	0.016611735	0.008765765	0.008020208	0.0003518438	0.08017421	0.006130423	0.0024461023	0.021123060	0.0021443050
A_{15}	0.013938217	0.041679621	0.016456935	0.019148257	0.009042695	0.007838123	0.0003329714	0.07785210	0.006125154	0.0028160545	0.022534021	0.0026753398
A_{16}	0.023694970	0.091366118	0.017379770	0.041740715	0.013619637	0.009473046	0.0003510216	0.09226437	0.008180172	0.0061557480	0.038423786	0.0075874116
A_{17}	0.011540844	0.035317995	0.013046159	0.016226283	0.007402205	0.006290503	0.0002639673	0.06235027	0.004958381	0.0023862841	0.018657736	0.0023190760
A_{18}	0.015833815	0.062958165	0.010676067	0.028772122	0.008933868	0.005981063	0.0002156718	0.05802114	0.005257869	0.0042424874	0.025670352	0.0053506796
A_{19}	0.016168332	0.051057536	0.016994047	0.023425531	0.010199538	0.008383121	0.0003437138	0.08279623	0.006710984	0.0034473661	0.026158160	0.0034405228
A_{20}	0.016335591	0.052209210	0.016851270	0.023935322	0.010251877	0.008350338	0.0003407816	0.08242545	0.006712068	0.0035237493	0.026438370	0.0035620783

Table 12. Weighted normalized decision matrix.

Alternatives	(B^+)	(B^-)	Alternatives	(B^+)	(B^-)
M1	0.0040699595	0.0276534233	RI	0.0001941686	0.0003518438
M2	0.0073487754	0.1276712615	SC	0.0922643727	0.0388520199
H	0.0095619559	0.0174069660	GA	0.0081801723	0.0026160824
F	0.0034193317	0.0581908084	HZI	0.0004997116	0.0085917272
SS	0.0033167282	0.0144775780	ReZ1	0.0448979475	0.0065408198
ABC	0.0037733235	0.0094730459	ReZ2	0.0125235704	0.0002845629

Table 13. Calculation of positive and negative ideal solution.

Alternatives	E_i^+	E_i^-	C_i	Rank
A_1	0.04518660	0.11122593	0.7111062	3.0
A_2	0.05480822	0.13161331	0.7059984	4.0
A_3	0.04901193	0.10884070	0.6895083	8.0
A_4	0.06683281	0.13279631	0.6652151	12.0
A_5	0.06077384	0.13542374	0.6902416	7.0
A_6	0.05069206	0.11809324	0.6996655	5.0
A_7	0.05609745	0.12860325	0.6962791	6.0
A_8	0.06355945	0.13026952	0.6720849	10.0
A_9	0.05396139	0.10421394	0.6588508	13.0
A_{10}	0.07434731	0.07788551	0.5116210	17.5
A_{11}	0.13564387	0.05833783	0.3007389	20.0
A_{12}	0.04501493	0.11094351	0.7113659	1.5
A_{13}	0.06833953	0.08609789	0.5574937	16.0
A_{14}	0.04501493	0.11094351	0.7113659	1.5
A_{15}	0.04934261	0.10472956	0.6797435	9.0
A_{16}	0.09597205	0.07459694	0.4373417	19.0
A_{17}	0.05233311	0.10665879	0.6708442	11.0
A_{18}	0.07434731	0.07788551	0.5116210	17.5
A_{19}	0.05581335	0.09797439	0.6370754	14.0
A_{20}	0.05687864	0.09676383	0.6297987	15.0

Table 14. Ranking to alternatives (drugs) via TOPSIS.

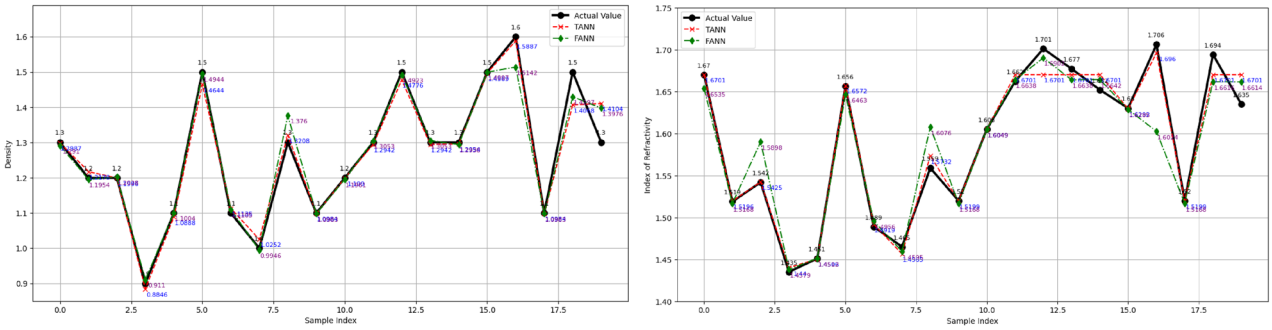


Fig. 9. Graphical comparison of density and index of refractivity.

Conclusion

In conclusion, this study demonstrates the successful application of artificial neural networks and fuzzy artificial neural networks algorithms for the prediction and assessment of potential anti-epileptic drugs. Both models performed effectively across various datasets, however the fuzzy artificial neural networks approach consistently provided more refined and accurate predictions due to its ability to handle uncertainty and incorporate

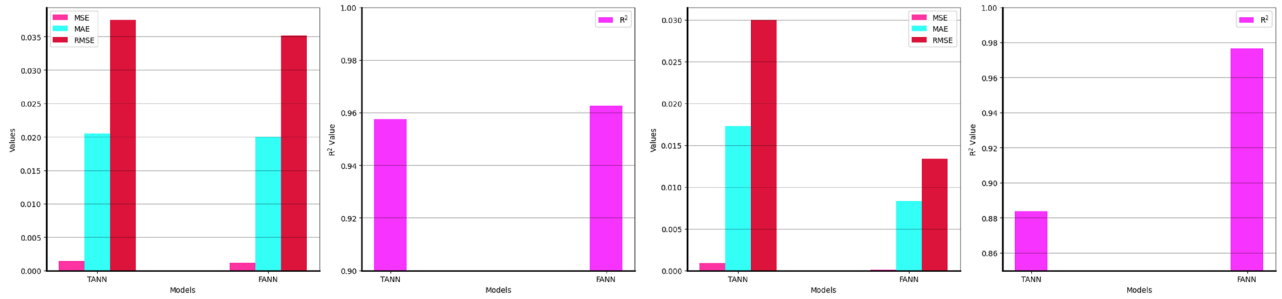


Fig. 10. Metrics comparison of density and index of refractivity.

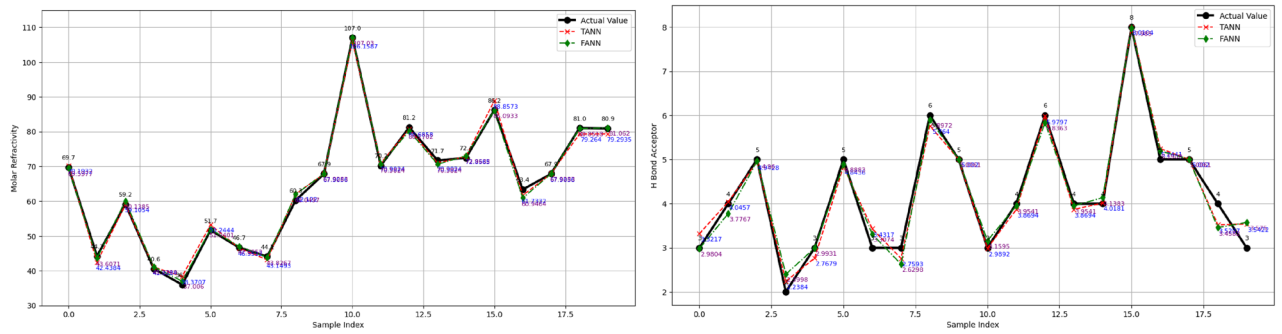


Fig. 11. Graphical comparison of molar refractivity and HBond acceptor.

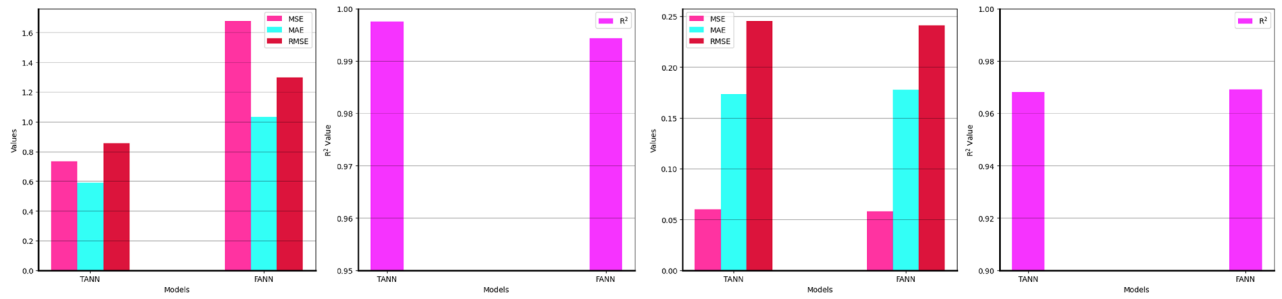


Fig. 12. Metrics comparison of molar refractivity and HBond acceptor.

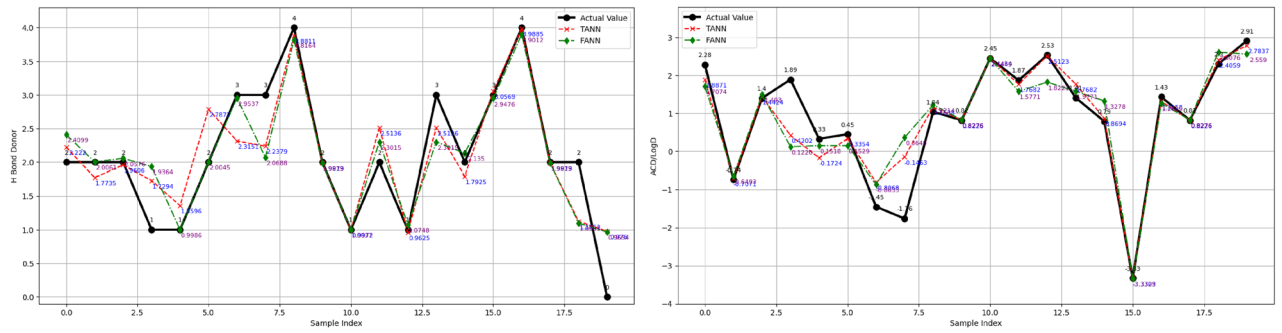


Fig. 13. Graphical comparison of HBond donor and ACD/LogD.

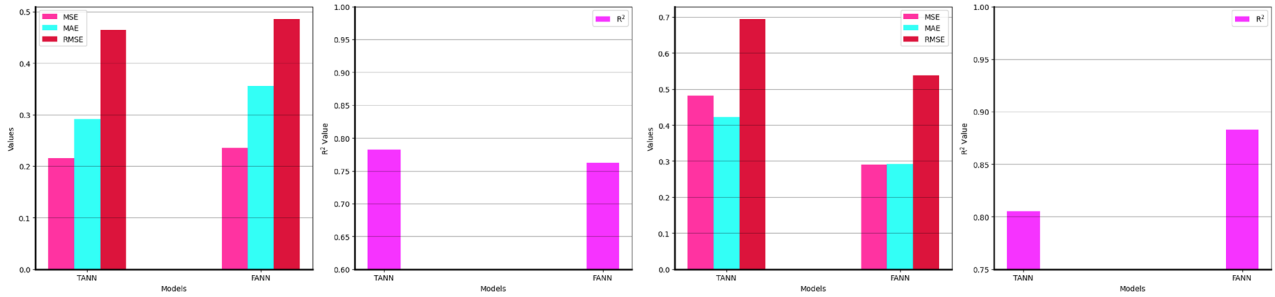


Fig. 14. Metrics comparison of HBond donor and ACD/LogD.

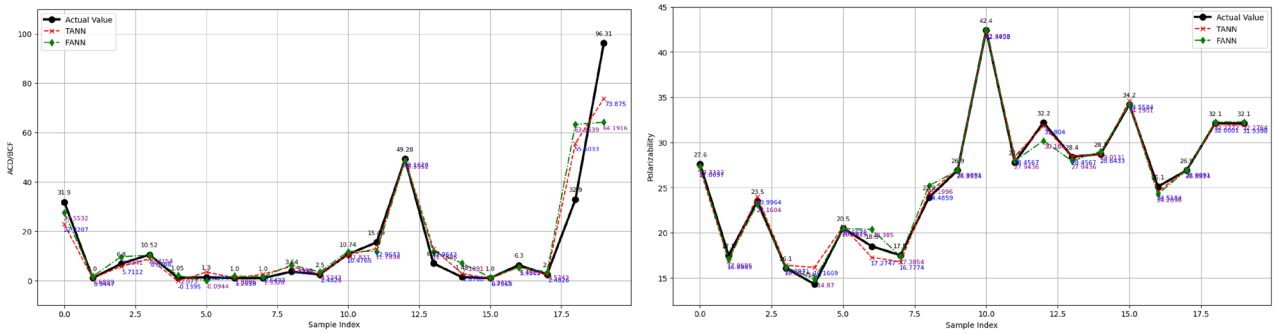


Fig. 15. Graphical comparison of ACD/BCF and polarizability.

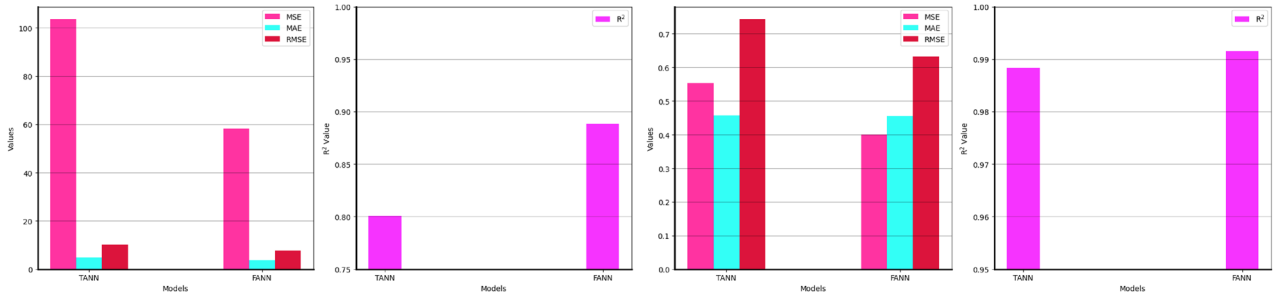


Fig. 16. Metrics comparison of ACD/BCF and polarizability.

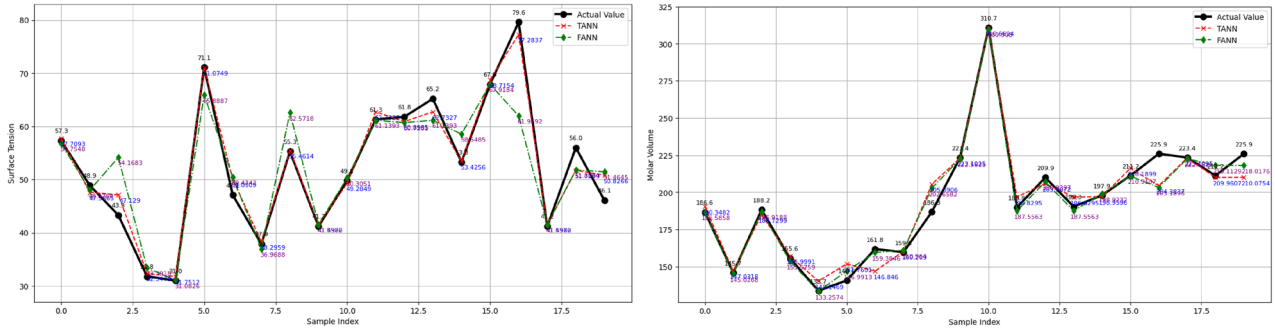


Fig. 17. Graphical comparison of surface tension and molar volume.

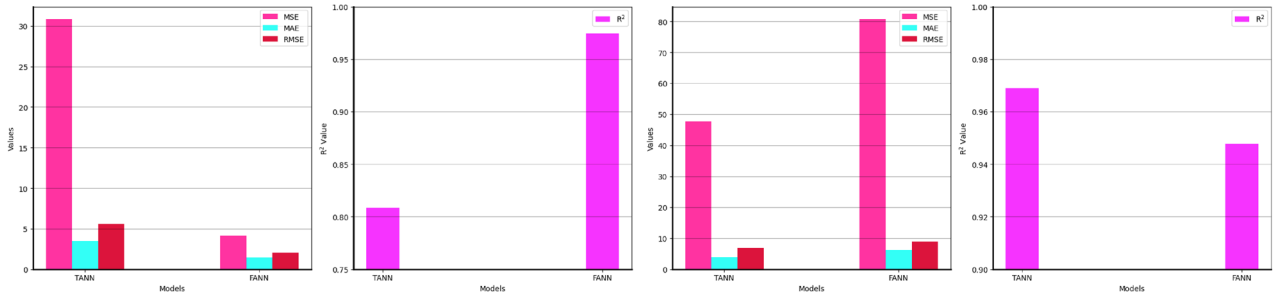


Fig. 18. Metrics comparison of surface tension and molar volume.

degrees of membership rather than binary classification. The results demonstrate that fuzzy-based approaches, especially FANNs, outperform most other algorithm as we see in Table 9 and have a higher predictive accuracy ($R^2 > 0.75$). This nuanced treatment of input features enables fuzzy artificial neural networks to better capture the complex, non-linear relationships present in biomedical data, particularly in pharmacological contexts. To further refine the selection of the most suitable drugs, we employed a multi-criteria decision-making method specifically, the technique for order of preference by similarity to ideal solution. TOPSIS enabled a systematic and objective evaluation of the drug candidates by considering multiple quantitative and qualitative factors such as predicted efficacy, safety profiles and computational model scores. Based on this comprehensive ranking system, Oxcarbazepine and Eslicarbazepine emerged as the top-ranking drugs, demonstrating optimal performance across the majority of selected criteria. The integration of machine learning models with fuzzy logic and decision-making frameworks like TOPSIS represents a powerful and intelligent approach for drug discovery and repositioning. This hybrid methodology not only enhances prediction robustness but also ensures that the final drug recommendations are both data driven and clinically relevant. Therefore, the findings of this study not only support the reliability of artificial neural networks and fuzzy artificial neural networks models but also highlight the importance of combining predictive analytics with decision support tools in the context of epilepsy treatment.

Limitations and future work

The proposed computational framework provides a robust approach for comparative evaluation and ranking of antiepileptic drug candidates, but several limitations should be acknowledged. To predict continuous physicochemical properties the study uses supervised regression models, including ANN and FANN. Regression specific measures (R^2, RMSE, MAE, MSE) were used to assess model performance, and no measures of classification were used since the study is not based on categorical results. The statistics power and external generalizability is limited by the relatively small size of the dataset of 20 drugs. In order to minimize overfitting and to improve the strength of the models, 5k fold cross-validation and regularization were implemented. Therefore, the findings can be seen as comparative and exploratory but not definitive.

Additionally, Despite the proposed modeling framework effectively describing the structure-activity relationships, it does not include explicit receptor level pharmacodynamics, mechanistic targeting information and drug interactions. These pharmacological dimensions play a vital role in defining target selectivity, off-target effects, as well as the synergistic or antagonistic interactions, especially in epilepsy, where combination therapy has become the common clinical procedure. The current model utilizes molecular topological indices, which represent global structural and physicochemical data but do not directly reflect ligand-receptor binding kinetics, modulation of signaling pathways, or interaction-induced pharmacodynamic variability. The further development of this work will focus on incorporating receptor-level descriptors (binding affinities, target occupancy measures), mechanistic pathway annotations, and rug interactions attributes based on pharmacokinetic and pharmacodynamic interaction networks. It is anticipated that through such integration, either through multimodal or graph-based learning, model interpretability, model robustness, and model translational relevance to clinically informed antiepileptic drug prioritization will be enhanced.

Data availability

All data generated or analysed during this study are included in this published article

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Declarations

Competing interests
The authors declare no competing interests.

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